

Dhana Shekhar

AI/ML Engineer | Data Scientist

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PROFESSIONAL SUMMARY

AI/ML Engineer with experience building production-oriented machine learning and GenAI solutions across FinTech, healthcare, insurance, and retail. Specializes in LLM applications, RAG, agentic workflows, document intelligence, APIs, and cloud-based ML deployment using Azure and AWS. Builds end-to-end solutions with evaluation, guardrails, observability, and secure integrations for regulated environments.

TECHNICAL SKILLS

Languages: Python, R, SQL, Java, C++

Data & ML Libraries: NumPy, Pandas, Scikit-learn, XGBoost, LightGBM, CatBoost, TensorFlow, PyTorch

ML Techniques: Predictive Modelling, Segmentation, Classification, Clustering

NLP/LLMs: BERT, GPT, OpenAI APIs, Hugging Face Transformers, LangChain, LangGraph, CrewAI, LangSmith, AutoGen, RAG, Model Context Protocol (MCP), Large Language Model Applications, Agentic AI, Context Engineering

Vector & Search: FAISS, Chroma, Milvus, Pinecone, FastText, Embeddings And Semantic Search

Prompt Engineering: Chain-of-Thought, Few-shot Prompting, Prompt Guardrails, N-gram Prompting

Time-Series Forecasting: ARIMA, SARIMA, Prophet, TFT

Testing & Explainability: SHAP, LIME, PDP, BLEU, ROUGE, AUC-ROC, RAGAS, Red Teaming

MLOps & CI/CD: MLflow, Docker, Kubernetes, Apache Airflow, Kedro, FastAPI, GitHub Actions, Azure DevOps

Cloud: Azure (OpenAI, AKS, AI Foundry), AWS (S3, Lambda, SageMaker, Glue), Serverless Architecture, Distributed Systems, Event-Driven Architectures

Security & Ethics: HIPAA, PII/PHI Redaction, Adversarial Prompt Testing

Visualization & Reporting: Power BI, Tableau, Matplotlib, Seaborn, Plotly

Databases: SQL, MongoDB, NoSQL, Snowflake

Version Control: Git, Bitbucket, TFS

Web Frameworks: Flask, Django, FastAPI, API Development And Service Integration

AI Tooling (Hands-on / POC): Claude Code, Cursor AI, GitHub Copilot, n8n, AI-Assisted Software Development Tools

PROFESSIONAL EXPERIENCE

National Datacare | *AI Engineer*

Apr 2025 - Present

Senior-Care FinTech Platform Modernization | Domain: FinTech | Cloud: Azure

- Designed and prototyped large language model applications for a senior-care compliance platform, applying LLM-based document parsing to regulatory documents and policy content; translated business requirements into Claude-assisted designs and wireframes, reducing manual compliance review time.
- Built a Product Assistant using LLMs, fine-tuning, retrieval-augmented generation, embeddings, and semantic search over product documentation, enabling contextual retrieval and grounded responses across product knowledge and operational documentation.
- Developed a Project Assistant for engineering workflows that combined infrastructure documentation, product requirements, source code, and GitHub repositories with Model Context Protocol-enabled tools and AI coding agents, including GitHub MCP, Cursor, Linear, and Claude Code, resulting in faster issue resolution and code generation.
- Built an AI incident co-pilot that ingests event-driven webhooks from source control, cloud monitoring, and security scanning through FastAPI, generates grounded incident narratives with an LLM, and serves results through a lightweight web portal; shipped with 33 automated tests and received leadership approval for production rollout.
- Implemented security and compliance controls across HIPAA, PCI DSS, and SOX-regulated workloads, including encryption in transit and at rest, tokenized payment-card data, audit-log retention and transaction-level change logging, and automated security scanning enforced as a mandatory pull-request merge gate.
- Built an AI bot orchestration capability for agentic AI services, establishing reusable patterns for tool calling, context exchange, human approval, and multi-step agent workflows across business and engineering use cases.
- Designed and prototyped an automated financial data acquisition service for thousands of custodial accounts, using API, SFTP, and browser automation with human-in-the-loop OTP and CAPTCHA handling, Entra ID authentication, and Azure Blob Storage, which cut data onboarding time in half
- Contributed AI and automation capabilities to core FinTech products such as Resident Admission and Invoice Management, identifying document-intelligence and workflow-automation opportunities that improved processing efficiency
- Developed an agent-assisted delivery workflow spanning ticket intake, analysis, implementation, security review, and pull-request creation, introducing mandatory human approval gates and evaluating MCP servers and AI coding agents to establish repeatable AI-assisted engineering patterns.
- Supported AI-ready modernization of the legacy Resident Fund Management System by defining service boundaries, data flows, and reusable platform capabilities, enabling migration to Azure-native services and laying groundwork for future AI products
- Established production engineering foundations for AI workloads using FastAPI, containerized services, AKS, GitOps, Bicep, GitHub Actions/OIDC, and Azure services, enabling AI applications and agents to move from prototypes toward governed, reliable deployment.

Cigna | *Machine Learning Engineer*

Jan 2022 - Dec 2023

Intelligent Healthcare Analytics & Patient Outcome Prediction | Domain: Healthcare & Life Sciences | Cloud: Azure

- Architected and deployed machine learning pipelines with Scikit-learn, XGBoost, LightGBM, TensorFlow, and PyTorch to predict patient outcomes, score risk, and detect early disease, improving clinicians' ability to intervene sooner
- Implemented Prophet, ARIMA, and TFT to analyze ICU vitals and longitudinal patient records, enabling predictive interventions and improved resource allocation.
- Established end-to-end workflows with MLflow, Docker, Kubernetes (AKS), and Apache Airflow, ensuring reproducibility, automated retraining, and scalable deployment.
- Automated Azure infrastructure provisioning with Terraform, reducing setup times and enabling rapid experimentation for healthcare ML workloads.
- Exposed ML models through API development and service integrations with FastAPI and hospital EHR systems, delivering real-time patient risk scores and alerts with attention to reliability and performance.
- Applied SHAP, LIME, and PDP to enhance transparency and support clinician trust and audit compliance.
- Enforced HIPAA safeguards, including PII/PHI redaction, adversarial prompt testing, and human-in-the-loop systems, supporting secure and ethical AI adoption.
- Developed ETL pipelines using Azure Data Factory, SQL, and Snowflake, consolidating heterogeneous medical datasets into structured, analysis-ready formats.
- Delivered interactive dashboards in Power BI and Plotly, enabling administrators to monitor patient outcomes and hospital KPIs.

CERTIFICATIONS

• Microsoft Certified: Azure AI Engineer Associate

• Oracle: Database Programming with SQL

EDUCATION

Master of Science, Management, Data Analytics | *Indiana Wesleyan University*

Jan 2024 - Aug 2025

Bachelor of Technology, Artificial Intelligence | *Vidya Jyoti Institute of Technology*

Aug 2019 - Jun 2022